

CS-010 CF Grounding & Electrical Bonding

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Title	Carbon Fiber Grounding & Electrical Bonding
Revision	C
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Effective date	2026-08-28
Supersedes	Tribal grounding knowledge from SN5 trials

Approvals

Role	Name	Date
Author	Simon Starbuck (drafted with Claude, SN6 Resources)	2026-07-12
Reviewer	simon (reviewer agent)	
Approver (Composites Lead)		

Revision history

Rev	Date	Author	Description of change
A	2026-07-12	Claude	Initial issue, encodes the diffuser-validated method + the catch-can decision lesson
B	2026-07-28	Claude	Language pass: fewer em dashes for rhythm variety, no content change.
C	2026-08-28	Claude	Engineering pass (CS-000 Rev C conventions). Figure 1 draws the diffuser recipe in plan: mesh sheets, the 25 mm overlap, the terminable tail, and the probe pattern that proves the part. One intensifier cut from §1. No rule changed.

1. Purpose

EV rules want conductive parts near HV/chassis grounded (our SN5 working criterion: **<5 Ω to the grounding point**). SN5 gave us both the success story and the cautionary tale. The **diffuser** (target and method written down up front, copper mesh in the stack) validated at **<1 Ω continuous everywhere** (2026-06-21) and drew praise from the design judges at comp. The **catch can** had no written criterion; its method was improvised for six months (Nov→May) and never hit target with CF wrapping, so we finally solved it by *giving up on CF for that geometry* and going to epoxy + copper paths (PP-05). This doc exists so we repeat the first one.

2. Scope

Any CF part with a grounding/bonding requirement, and any conductive-path retrofit on existing parts. The Ω target and rule interpretation come from the EV rules owner (powertrain/rules lead); this standard covers how composites *achieves and proves* the target, not what the target legally is.

3. References

Ref	Document	Where
R1	PP-05 RCA (catch-can saga)	FEB-COMP-PP-001
R2	CS-002 (mesh in the ply table)	CS-002 §7.1
R3	SN5 grounding trials: graphite powder judged insufficient vs copper mesh (Nico, #composites, Nov 2025)	Slack record
R4	Diffuser validation $<1\ \Omega$ (post-comp recap)	#composites, 2026-06-21

4. Definitions

- **Grounding point:** the fastener/tab where the part's conductive path meets chassis ground.
- **Probe pattern:** the defined set of surface points measured against the grounding point.
- **CF as conductor:** carbon laminate conducts, but through-thickness resistance through resin-rich layers is high and unpredictable; bare CF also can't be reliably terminated. Hence copper mesh.

5. Equipment & Materials

Item	Spec / product	Source	Notes
Copper mesh	Fine woven mesh (repeat team buy)	Amazon	The validated conductor (R3, R4)
Multimeter	2-wire, known-good leads	Electrical	Zero the leads first (touch together, note offset)
Ring terminals / grounding hardware	Per part design	McMaster	
Graphite powder	1–5 wt% in resin	Sigma-Aldrich	Secondary/supplement only; insufficient alone (R3)

6. Safety

Grounding verification happens on or near the car: follow EV safety rules. No probing near energized HV; HV isolation confirmed by the ESO before resistance measurements on installed parts. Bench measurements on loose parts: no special hazards beyond CS-009 dust rules if sanding contact points.

7. Procedure

7.1 Decision point first (the catch-can lesson)

1. Before any layup work, answer on the WO: **is CF-with-copper-mesh the right conductive path for this geometry?**

- **Good fit:** large open laminates (diffuser, panels) where mesh lays flat in the stack and terminates at accessible flange points.
 - **Bad fit:** thin-walled, tapered, or wrapped metal substrates (the catch can: sanding punctured the can wall at the taper; through-thickness contact never stabilized). For these, use **epoxy + copper strip/braid paths bonded to the surface**, or reconsider whether the part needs CF at all.
2. Write the acceptance criterion on the WO **before work starts:** Ω target, probe pattern (grid spacing or named points), and measurement method (7.3). Six months of drift happened because none of this was written (R1).

7.2 Build the path (mesh-in-stack method: the diffuser recipe)

1. Put copper mesh in the ply table per CS-002: mesh layer(s) positioned to be continuous from part surface zones to the grounding point(s), **25 mm minimum overlap** where mesh sheets meet.
2. Mesh must reach the grounding point as bare, terminable material: leave a mesh tail at the flange/fastener location, protected during infusion (peel-ply window or masked zone).
3. For surfaces needing exposed conductivity, an 88-gsm outer ply over mesh keeps finish while staying thin; validate on the first article that surface-to-ground still passes.
4. Graphite powder (1–5 wt% in the resin) may be added as a *supplement* in zones where mesh coverage is marginal, never as the primary path (SN5 trials judged it insufficient alone, R3). If used, record wt% on the WO.
5. Infuse/laminate per CS-006/CS-007. Note on the WO where mesh actually ended up (as-built).

Plan view: mesh-in-stack path (the diffuser recipe, §7.2) and the §7.3 probe pattern

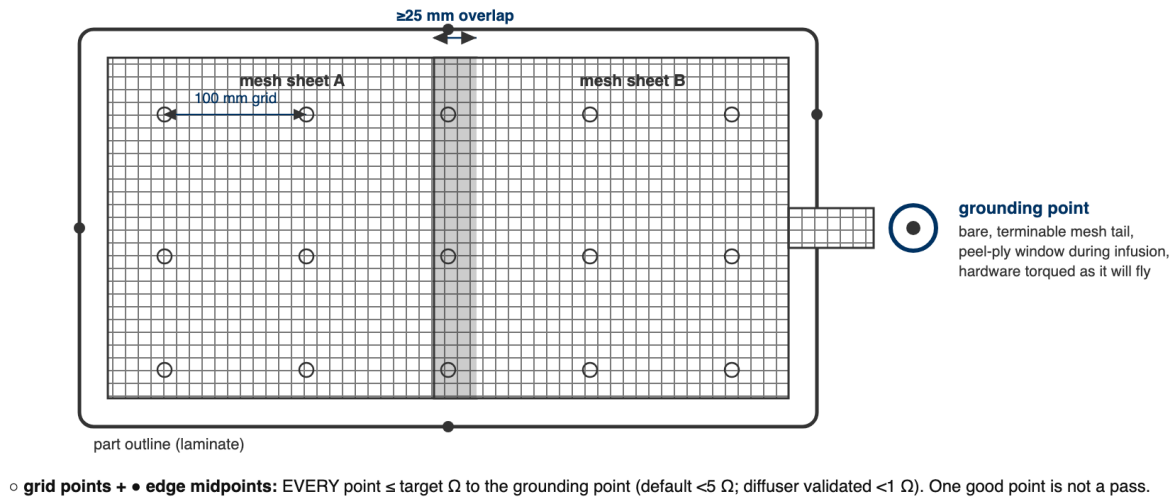


Figure 1: The mesh-in-stack path in plan: sheets overlapping 25 mm, a bare terminable tail at the grounding point, and the default probe pattern (100 mm grid plus edge midpoints). Every point measures to the grounding point; every point passes or the part does not.

[PHOTO: copper mesh tail at grounding point before bagging]

7.3 Measure: one method, every time

1. Zero the multimeter leads; record lead resistance.
2. Fasten the grounding hardware as it will fly (torqued, washers as designed); hand-held contact readings are noise.

3. Probe every point in the WO's probe pattern (default: 100 mm grid over conductive zones + every edge midpoint) to the grounding point. Record each value minus lead offset.
4. Pass = **every** point target (default $<5\ \Omega$, or the rules value on the WO). Not "one good point": that ambiguity burned a month in May 2026 (R1).

7.4 If it fails

1. Map the failing zone; usually a mesh discontinuity or resin-isolated overlap.
2. Fixes in order of preference: expose + bond a copper strip bridge (epoxy + copper path), add an external ground tab at the failing zone, or lightly abrade to mesh at a termination point and re-terminate. **Do not sand thin laminates chasing contact.** That is how the catch cans died (R1).
3. Re-measure the full pattern after any fix, not just the failed point.

8. Quality checks / acceptance criteria

Check	Target	Method	Record where
Criterion written before work	Ω + probe pattern + method on WO	WO blocker step	WO
Decision point answered	CF-mesh vs alternative path, with one-line why	WO field	WO
Resistance	Every probe point target	§7.3 method	WO qualityChecks (full table of readings)
As-built mesh record	Matches or deviations noted	CS-002 §7.4	WO

9. Records

The WO keeps: the decision-point answer, criterion, full probe-reading table (values, not "pass"), photos of mesh placement and termination, and any retrofit fixes. The diffuser's $<1\ \Omega$ table is the reference example once WO-SN5 retro-records are loaded.